

Class 7 — Set A

Mateng EduFest 2026

1. Which one is true?

- A) Two acute angles can be complementary to each other.
- B) Two obtuse angles can be complementary to each other.
- C) Two right angles can be complementary to each other.
- D) None of these is true.

2. Two congruent triangles have:

- A) Equal area
- B) Equal perimeter
- C) Both A and B
- D) None of these

3. The median of the data 24, 36, 46, 17, 18, 25, 35 is

- A) 25
- B) $28 \frac{4}{7}$
- C) 17
- D) None of the above

4. If two lines intersect at a point, then the vertically opposite angles are always ____.

- A) 0°
- B) Equal
- C) 180°
- D) Straight line

5. Which one is correct?

- A) A proper fraction has numerator greater than denominator.
- B) Every mixed fraction is a proper fraction.
- C) $\frac{1}{5}$ can be considered an improper fraction.
- D) An improper fraction can be converted into a mixed fraction.

6. The value of n in $2n - 1 = \frac{27}{5}$ is:

- A) 6
- B) 3.5
- C) 5.4
- D) 3.2

7. A car covers a distance of 89.1 km in 2.2 hours. What is the average distance covered by it in 1 hour?

- A) 405 km
- B) 40 km 5 m
- C) 40 km 500 m
- D) 40.5 m

8. Chalk contains calcium, carbon and oxygen in the ratio 10 : 3 : 12. Find the percentage of carbon in chalk.

- A) 40%
- B) 12%
- C) 48%
- D) 30%

9. Saili plants 4 saplings in a row in her garden. The distance between two adjacent saplings is $\frac{3}{4}$ m. Find the distance between the first and the last sapling.

- A) 3 m
- B) $\frac{9}{4}$ m
- C) $\frac{6}{4}$ m
- D) $\frac{3}{4}$ m

10. Which of the following is true?

- A) A triangle has only one median.
- B) There is only one altitude of a triangle.
- C) The median of a triangle joins a vertex and the mid-point of the opposite side.
- D) The sum of all angles of a triangle is 360° .

11. Explain why $\triangle ABC \cong \triangle FED$.

- A) RHS congruence
- B) ASA congruence
- C) SAS congruence
- D) SAA congruence

12. $3.5 = \underline{\hspace{1cm}}\%$

- A) 35
- B) 0.35
- C) 350
- D) 0.035

13. Which one has the highest value?

- A) $0.5 \times \frac{1}{5}$
- B) $\frac{2}{5} \div 2$
- C) 3×0.3
- D) $1 - \frac{3}{10}$

14. The equivalent of 2.5 hr is

- A) 2 hr 50 min
- B) 2 hr 30 min
- C) 2 hr 5 min
- D) 250 min

15. Which one is incorrect?

- A) Sum of a number and its additive inverse is zero.
- B) Product of a number and its multiplicative inverse is 1.
- C) 1 is the additive identity.
- D) Product of a number and 0 is always 0.

16. The mean of the first five whole numbers is

- A) 3
- B) 2
- C) 1
- D) 0

17. A man travelled two-fifths of his journey by train, one-third by bus, one-fourth by car, and the remaining 3 km on foot. What is the length of his total journey?

- A) 180 km
- B) 100 km
- C) 200 km
- D) 30 km

18. $625 \times (-35) + (-625) \times 65 =$

- A) 62500
- B) -18750
- C) 18750
- D) -62500

19. Which is the least popular pet, according to the bar graph?

- A) Dogs
- B) Cats
- C) Rabbits
- D) Hamsters

20. The value of x is

- A) 35°
- B) 90°
- C) 125°
- D) Cannot determine

21. Choose the incorrect option:

- A) Sum of two whole numbers is a whole number.
- B) Product of two integers is an integer.
- C) The difference between two rational numbers is always a rational number.
- D) $22/7$ is not a rational number.

22. At Srinagar, the temperature was -5°C on Monday and then it dropped by 2°C on Tuesday. On Wednesday, it rose by 9°C . The temperature on Wednesday is ____ $^\circ\text{C}$ more than the temperature on Monday.

- A) 4
- B) 7
- C) 2
- D) 6

23. The observation that occurs most often in a given data is the

- A) Mean
- B) Median
- C) Mode
- D) Frequency

24. In a class test containing 10 questions, 5 marks are awarded for every correct answer, -2 marks for every incorrect answer, and 0 for questions not attempted. Reshma gets five correct answers and five incorrect answers. What is her score?

- A) 15
- B) 12
- C) 25
- D) 50

25. Which of the following is true?

- A) All rational numbers are natural numbers.
- B) For any given rational number, we can find its immediate adjacent rational number.
- C) 0 is not a rational number.
- D) All integers are rational numbers.

26. Ratio of 30 days to 36 hours

- A) 10 : 12
- B) 20 : 1
- C) 12 : 10
- D) 1 : 20

27. Reena's mother said, "To make idlis, you must take three parts rice and two parts urad dal." What percentage of such a mixture would be rice?

- A) 40%
- B) 60%
- C) 80%
- D) 100%

28. Which one is the least?

- A) $\frac{2}{9}$
- B) $\frac{2}{3}$
- C) $\frac{8}{21}$
- D) $\frac{1}{2}$

29. If $l \parallel m$, the value of x is

- A) 70°
- B) 110°
- C) 60°
- D) Cannot determine

30. The average of 4.2, 3.8 and 7.6 is

- A) 5.2
- B) 3.8
- C) 4.2
- D) 7.6

31. An elevator descends into a mine shaft at the rate of 6 m/min. If the descent starts from 30 m above ground level, how long will it take to reach -360 m?

- A) 55 min
- B) 1 hr
- C) 1.5 hrs
- D) 1 hr 5 min

32. Which of the following is not a congruence criterion of triangles?

- A) ASA congruence rule
- B) SSS congruence rule
- C) RHS congruence rule
- D) AAA congruence rule

33. In a right triangle, the longest side is 13. If one side is 5, then the ratio of the other remaining side to the hypotenuse is

- A) 12 : 13
- B) 5 : 13
- C) 12 : 5
- D) 12 : 18

34. In a pair of adjacent angles: (i) the vertex is always common, (ii) one arm is always common, and (iii) the uncommon arms are always opposite rays. Then:

- A) All of (i), (ii) and (iii) are true
- B) (iii) is false
- C) (i) is false but (ii) and (iii) are true
- D) (ii) is false

35. Which one is true?

- A) A triangle in which two sides are of equal length is called a scalene triangle.
- B) The sum of the lengths of any two sides of a triangle is less than the third side.
- C) If the Pythagoras property holds, the triangle must be right-angled.
- D) The difference between the lengths of any two sides is greater than the length of the third side.

36. Choose the incorrect one:

- A) Supplementary angle of 100° is 80° .
- B) Complementary angle of 90° is 0° .
- C) Adjacent angles have a common arm.
- D) Linear pair angles are not adjacent angles.

37. Sameera purchased $3\frac{1}{2}$ kg apples and $4\frac{3}{4}$ kg oranges. What is the total weight of fruits purchased by her?

- A) $7\frac{4}{6}$ kg
- B) $7\frac{2}{3}$ kg
- C) $8\frac{1}{4}$ kg
- D) Both A and B

38. Which of the following cannot be the sides of a triangle?

- A) 2.5 cm, 6.5 cm, 6 cm
- B) 2 cm, 2 cm, 5 cm
- C) 1.5 cm, 2 cm, 2.5 cm
- D) None of the above

39. If A is increased by 20%, it equals B. If B is decreased by 50%, it equals C. Then ____% of A is equal to C.

- A) 70
- B) 60
- C) 50
- D) 20

40. An article was sold for ₹250 with a profit of 25%. What was its cost price?

- A) ₹200
- B) ₹625
- C) ₹1000
- D) None of these

41. In general, for any integers a , b and c , we can say $a + (b + c) = (a + b) + c$. Which property is this?

- A) Closure property
- B) Commutative property
- C) Associative property
- D) Distributive property

42. The angle which is equal to its supplement is

- A) 0°
- B) 45°
- C) 90°
- D) 180°

43. Select the correct option:

- A) Commutative property holds for both addition and multiplication of integers.
- B) Product of two negative integers is less than 0.
- C) 0 is a positive integer.
- D) -10 is more than -5 .

44. ₹10,000 is invested at 5% interest rate p.a. Find the interest at the end of 6 months.

- A) ₹3000
- B) ₹500
- C) ₹250
- D) ₹1500

45. The ages, in years, of 10 teachers of a school are: 32, 41, 28, 54, 35, 26, 23, 33, 38, 40. The range is

- A) 54
- B) 31
- C) 35
- D) 23

Questions 46–50

Questions 46–50: There are two or more correct options.

46. Equivalent fractions of $\frac{22}{7}$ are

- A) π
- B) $\frac{44}{14}$
- C) $\frac{110}{35}$
- D) 3.14

47. Which of the following are true?

- A) Two right angles of different arm lengths are not congruent angles.
- B) Two right triangles are always congruent.
- C) There can be at most one right angle in a triangle.
- D) An obtuse triangle has only one obtuse angle.

48. Select the correct options:

- A) You are older today than yesterday is certain.
- B) A tossed coin will land heads up is not certain.
- C) A die, when tossed, shall land up with 8 on top is impossible.
- D) The next traffic light seen will be pink is certain.

49. If $l \parallel m$, then

- A) $\angle 1$ and $\angle 8$ are supplementary.
- B) $\angle 2$ and $\angle 4$ are a linear pair.
- C) $\angle 3 = \angle 8$
- D) $\angle 4 = \angle 7$

50. Identify the correct formula (symbols have their standard meaning):

- A) Simple Interest = $(P \times R \times T) / 100$
- B) Decimal $\times 100$ = Percentage value of the decimal
- C) Selling Price = Cost Price + Profit %
- D) Simple Interest = Amount – Principal